

# Notes on Distributions, Sobolev Spaces and Weak Solutions

Written by: Caio Giroldo Valério

The main objective of this text is to compile everything I've read or discussed about weak solutions on PDEs and the prerequisites for understanding this topic.

---

# Summary

|          |                                                                            |          |
|----------|----------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Motivation</b>                                                          | <b>3</b> |
| <b>2</b> | <b>Test functions and Distributions</b>                                    | <b>5</b> |
| 2.1      | The elements of $\mathcal{D}(\Omega)$ and $\mathcal{D}'(\Omega)$ . . . . . | 5        |
| 2.2      | Distributional derivatives . . . . .                                       | 6        |

---

---

# CHAPTER 1

---

## Motivation

We will begin by studying the solution of the following equation:

$$u_t + uu_x = 0, \text{ for } x \in \mathbb{R} \text{ and } t > 0 \quad (1)$$

This is commonly known as the inviscid Burger's equation, a simplified model for fluid mechanics, modelling one dimensional flow where  $u(x, t)$  is the velocity of the particle located in  $x$  at time  $t$ . So, first, we are going to try to find regular solutions for this equation by the method of characteristics.

Considering  $u(x, t)$  to be a smooth solution for (1) satisfying the initial condition problem:

$$u(x, 0) = u_0(x)$$

Where  $u_0(x)$  is a function of  $x$ . We'll start by defining the characteristic curves:

$$\begin{cases} \dot{x}(t) = u(x(t), t), & x(0) = x_0 \\ \dot{y}(t) = 1, & y(0) = y_0 \end{cases}$$

Solely from that, we get  $y = t$ . And now, by calculating the total derivative of  $u$  on the curves just defined, we obtain:

$$\frac{du}{dt} = \frac{\partial u}{\partial t} \frac{\partial y}{\partial t} + \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} = u_t + u_x u = 0$$

From this, two conclusions are made:  $u$  is constant along the curve  $(x(t), t)$ , which implies that  $\dot{x}(t) = u_0(x_0)$ . By solving this last ODE, we conclude that:

$$x = x_0 + u_0(x_0)t$$

And for all points in this curve  $u(x, t) = u_0(x_0)$ . For convenience's sake, we will choose this as our initial curve  $u_0$ :

Using the initial condition given by  $u_0$ , the characteristic curves are:

We can clearly see that there are curves intersecting each other, so the value of  $u(x, t)$  is not well-defined at the intersection points, therefore this cannot be a solution for our equation.

Now, the question that rises is: "if that isn't a solution, then what is?". And that will motivate our next few chapters of functional analysis definitions and theorems, aiming to build a generalized perspective on what is a solution for a PDE.

---

---

# CHAPTER 2

---

## Test functions and Distributions

Our main objective in this chapter is to make useful definitions to generalize the derivative of a function. For that, we will need a mechanism that can weaken the conditions for a function to have partial derivatives, by removing it entirely and passing it to another convenient function that always has a derivative in any axis.

### 2.1 The elements of $\mathcal{D}(\Omega)$ and $\mathcal{D}'(\Omega)$

#### Definition 2.1: Test functions

Let  $\Omega$  be a subset of  $\mathbb{R}^n$ . Then  $\mathcal{D}(\Omega)$  is the set of all  $C^\infty$  functions defined on  $\Omega$  with compact support.

An example of such function is:

$$f(x) = \begin{cases} \exp(\frac{-a^2}{a^2-x^2}), & |x| < a \\ 0, & |x| \geq a \end{cases}$$

And, for any  $\varphi$  on this set and any locally integrable function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ , using integration by parts we get:

$$\int_K D^\alpha f \varphi = (-1)^{|\alpha|} \int_K f D^\alpha \varphi \quad (2)$$

Where  $K \supseteq \text{supp} \varphi$  is a compact subset of  $\Omega$  and  $\alpha$  is a multi-index. This will motivate our next definition, but before that, we need to quickly describe the topology of  $\mathcal{D}(\Omega)$ :

A sequence  $\{\varphi_n\}_{n \in \mathbb{N}}$  in  $\mathcal{D}(\Omega)$  converges to  $\varphi$  if there exists a compact set  $K \subset \Omega$  such that  $\text{supp}(\varphi_n) \subset K$  for all  $n$  and  $D^\alpha \varphi_n \rightarrow D^\alpha \varphi$  for all  $\alpha$ .

Now we can define the most important tool used in this endeavour:

**Definition 2.2: Distributions**

A linear functional  $T$  is said to be a distribution if for all sequences  $\{\varphi_n\}_{n \in \mathbb{N}}$  in  $\mathcal{D}(\Omega)$  convergent to  $\varphi$ , the sequence  $\{T\varphi_n\}_{n \in \mathbb{N}}$  converges to  $T\varphi$  in  $\mathbb{R}$ .

We will use  $\mathcal{D}'(\Omega)$  for the set of all distributions. Note that  $\mathcal{D}'(\Omega)$  uses the same notation as the dual space of  $\mathcal{D}(\Omega)$  and that is no coincidence, as the distributions are precisely the dual elements of the test functions, as they are the linear functionals defined from  $\mathcal{D}(\Omega)$  to  $\mathbb{R}$ . Furthermore, it is worth to point that both  $\mathcal{D}(\Omega)$  and  $\mathcal{D}'(\Omega)$  are vector spaces over  $\mathbb{R}$ .

Now, aiming for (2), we will define the following:

**Definition 2.3: Distribution given by a function**

Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be a locally integrable function, we say that the distribution given by  $f$  is:

$$T_f(\varphi) = \int_{\Omega} f\varphi$$

It is trivial to verify that  $T_f$  is in fact a distribution, because as  $\varphi_n \rightarrow \varphi$ ,  $\int_{\Omega} f\varphi_n \rightarrow \int_{\Omega} f\varphi$ .

Another useful distribution is the Dirac "δ-function", which is in fact a distribution given by:

**Definition 2.4: The Dirac distribution**

For any  $\varphi \in \mathcal{D}(\Omega)$ :

$$\delta(\varphi) = \varphi(0)$$

These both distributions will be essential for the next sections.

## 2.2 Distributional derivatives

Having (2) in mind, let's build a definition of a distribution derivative that can satisfy that equation. Let  $f$  be a smooth function in  $\Omega$ , so:

$$D^{\alpha}Tf(\varphi) = D^{\alpha} \int_{\Omega} f\varphi$$

Now, by the dominated convergence theorem we can push the derivative inside the integral, that is:

$$D^{\alpha} \int_{\Omega} f\varphi = \int_{\Omega} D^{\alpha}f\varphi = (-1)^{|\alpha|} \int_{\Omega} f D^{\alpha}\varphi$$

So, we now know what we have to define  $D^{\alpha}T$  as to serve our needs. We let  $f$  as a smooth function just so  $D^{\alpha}f$  has some meaning in a sense, but in general, the only condition  $f$  must abide is to be locally integrable.

**Definition 2.5: Derivative of a distribution**

Let  $T \in \mathcal{D}'(\Omega)$ , the  $\alpha$  derivative of  $T$  is:

$$D^\alpha T(\varphi) = (-1)^{|\alpha|} T(D^\alpha \varphi)$$

Another important thing to note is that, for a distribution given by a function  $f$ , the following identity is verified:

**Lemma 2.1**

$$D^\alpha T_f(\varphi) = T_{D^\alpha f} = (-1) T_f(D^\alpha \varphi)$$

Which was already proven before. In addition to that, we will say that a locally integrable function  $f$  has a distributional derivative if there exists a distribution  $G$  such that for all  $\varphi \in \mathcal{D}(\Omega)$ :

$$\frac{d}{dt} T_f(\varphi) = G(\varphi)$$

So  $G$  is the distributional derivative of  $f$ . If  $G = T_g$  for some locally integrable function  $g$ , then we also say that  $g$  is the distributional derivative of  $f$ . This distributional derivative is also what is called the weak derivative of  $f$ , sometimes denoted as  $\partial_t f$ .

It should be clear that if a function has a regular derivative, its weak derivative will be the same, as the identity above will be satisfied by:

$$\frac{d}{dt} T_f = T_{\frac{df}{dt}}$$

**Example 2.1: The derivative of the Heaviside Function**

The Heaviside function is defined as:

$$H(x) \begin{cases} 1, x \geq 0 \\ 0, x < 0 \end{cases}$$

So the distributional derivative of  $H$  is the  $\delta$  distribution.

*Proof.* Let  $\Omega \in \mathbb{R}$ . As  $H$  is a locally integrable function, let's evaluate the distribution given by  $H$ :

$$T_H(\varphi) = \int_{\Omega} H(x) \varphi(x) dx$$

So, by taking the derivative of this distribution and using lemma 2.1 we get:

$$\frac{d}{dx} T_H(\varphi) = -T_H\left(\frac{d\varphi}{dx}\right) = - \int_0^{+\infty} H(x) \frac{d\varphi}{dx} dx = \varphi(x) \Big|_0^{+\infty} = \varphi(0) = \delta(\varphi)$$

Therefore,  $\delta$  is the distributional derivative of the Heaviside function. ■